INCH-POUND

MIL-C-24643/32D
22 November 1994
SUPERSEDING
MIL-C-24643/32C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWAU (INCLUDING VARIATION LS2SWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS2SWAU (unarmored), LS2SWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

	First	- Copper conductor, uncoated, ASTM B 286, size 22-7. maximum diameter 0.033 inch.
	Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
	Third	- Crosslinked polyethylene insulation. Nominal wall thickness 0.013 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
*	Fourth	- Two conductors cabled together with a nominal lay of 2.5 inches to form a pair.
	Fifth	- Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
*	Sixth	- Braided shield of AWG no. 34 or no. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
	Seventh	- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (watertight) (continued)

Eighth - The required number of pairs (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used to make a firm, well rounded assembly.
Ninth - Binder tape, applied helically with overlap.

Construction variant for type LS2SWAU

Tenth - Cross-linked polyolefin jacket (see table I for thickness).

Construction variant for type LS2SWA

Tenth - Cross-linked polyolefin jacket (See table I for thickness).
Eleventh - Braided metal armor.

TABLE I. Details.

Military 1/ part no. M24643/32	Size 2/	Number of pairs	Cable jacket thickness min avg (inch)	Overall diameter (inches)			
				TYPE LS2SWAU		TYPE LS2SWA	
				min	max	min	max
-01UN	3	3	0.050	0.480	0.520	0.530	0.570
-02UN	7	7	.050	.610	.660	.660	.710
-03UN	10	10	.050	.770	.830	.820	.880
-04UN	14	14	.063	.860	.930	.910	.980
-05UN	19	19	.063	.970	1.040	1.020	1.090
-06UN	24	24	.063	1.120	1.210	1.170	1.260
-07UN	30	30	.063	1.190	1.280	1.240	1.330
-08UN	37	37	.063	1.290	1.380	1.340	1.430
-09UN	44	44	.063	1.460	1.550	1.510	1.600
-10UN	61	61	.075	1.660	1.740	1.710	1.790

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS2SWAU and LS2SWA.

EXAMINATION AND TESTS:

Basic electrical:

Conductor Resistance - ohms/1000 feet at 25°C, maximum.....	17.37
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Shield to shield (pair shields only)	500
Insulation resistance - Megohms/1000 feet minimum	
Conductor to conductor	500
Component shield to shield	100
Conductor and shield continuity	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum	1.0
	Requirements
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum	30
Capacitance unbalance - percent maximum	8
Characteristic impedance - at 1MHz, ohms	75 ± 5
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Drip - 95 ± 1°C	Zero
Attenuation - at 3 MHz, decibels per 100 FT, maximum	3
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Pulse response time - nanoseconds/500 FT, maximum	400
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
Cable sealant removability	No failure
Armor (type LS2SWA only) - conformance to material, construction and coverage	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Shield - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 30	1000
37 through 61	500

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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